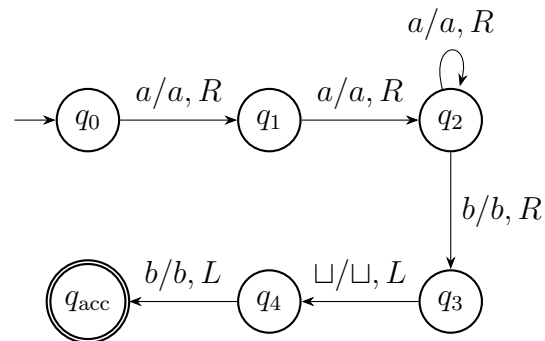


Quiz 2 — April 28, 2026

CPSC 406 — Algorithm Analysis,
Chapman University · Spring 2026
Instructor: Jonathan Weinberger

Name: _____

1. (2 points) (a) Consider the following deterministic Turing machine over input alphabet $\Sigma = \{a, b\}$. The blank symbol is $\sqcup$. All omitted transitions go to a rejecting halt state.



Which of the following words is accepted by the machine? Tick exactly one.

- ☐ abb
☐ aab
☐ $aabb$
☐ All of the above.
☐ None of the above.

(b) Let

$$f(n) = 7n^2 \log n + 3n + 10.$$

Which of the following Big-O classes contain $f(n)$? Tick exactly one.

- ☐ $\mathcal{O}(n \log n)$
☐ $\mathcal{O}(n^2)$
☐ $\mathcal{O}(\exp(n))$
☐ All of the above.
☐ None of the above.

2. (2 points) Show that the following formula, given in CNF, is unsatisfiable using resolution:

$$\varphi = (a \vee b \vee c) \wedge (\neg a \vee d) \wedge (\neg b \vee d) \wedge (\neg c \vee d) \wedge (\neg d \vee e \vee f) \wedge a \wedge \neg e \wedge \neg f.$$